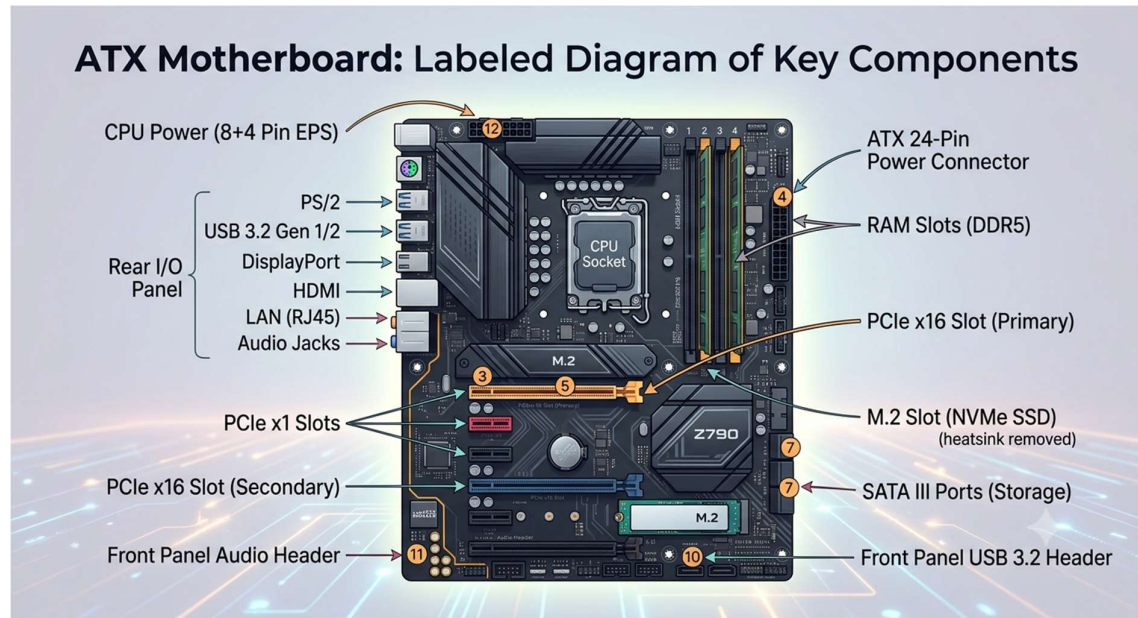


1. Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



2. Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

1. AMD Socket AM5

- **The Breakdown:** This is AMD's flagship desktop platform. It transitioned AMD away from pins on the CPU to flat contact pads (LGA layout), exclusively utilizing fast DDR5 memory and PCIe 5.0 lanes.
- **Why it's popular:** It is highly future-proof. AMD has committed to supporting this exact socket for multiple processor generations, allowing easy upgrades down the line without swapping the motherboard.
- ★ **Compatible CPU:** AMD Ryzen 7 7800X3D (or the Ryzen 7 9700X).

2. Intel LGA 1851

- **The Breakdown:** This socket is Intel's recent desktop platform design, built for their modular chiplet architecture (such as the "Arrow Lake" processors).
- **Why it's popular:** It focuses entirely on modern component architectures, cutting out older DDR4 support completely in favor of high-bandwidth DDR5 RAM and advanced onboard AI-processing capabilities.
- ★ **Compatible CPU:** Intel Core Ultra 7 265K (or the Core Ultra 9 285K).

3. Intel LGA 1700

- **The Breakdown:** This long-running platform supports three major generations of Intel processors (12th, 13th, and 14th Gen Core CPUs).
 - **Why it's popular:** Because it is a mature, well-established platform, it provides exceptional budget value. It uniquely supports both cheaper DDR4 or premium DDR5 memory layouts depending on the motherboard chosen.
 -  **Compatible CPU: Intel Core i7-14700K** (or the budget-friendly **Core i5-13400F**).
3. Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

1. The CPU: The Brains of the Operation

The **CPU (Central Processing Unit)** is the master controller. It doesn't store files or handle fancy graphics on its own; instead, it performs ultra-fast calculations and follows instructions. Every time you click a button, load a webpage, or mute a video, the CPU is processing the mathematical code behind that action.

2. The Chipset: The Traffic Cop

The **Chipset** is a crucial set of silicon chips built into the motherboard. If the CPU is the brain, the chipset is the **nervous system**. It acts as a smart traffic cop that directs data flowing between the CPU and all your other hardware—like your storage drives, RAM, USB ports, and audio hardware. The CPU is too busy thinking to talk directly to every little plug and port, so the chipset manages those conversations for it.

Behind the Scenes: Opening Spotify

When you double-click the Spotify icon, the CPU and chipset immediately team up to get the music playing. Here is how they work together step-by-step:

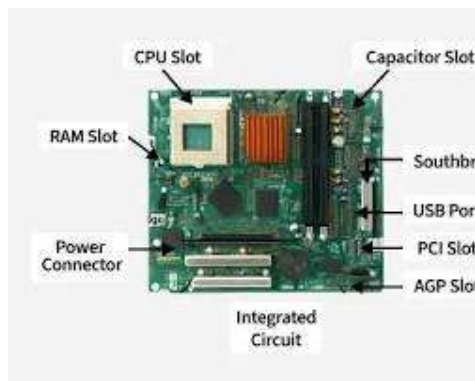
[Your Click] → (Chipset) → [CPU Processes Command] → (Chipset) → [SSD Loads Data] → [Music Plays!]

- **Step 1: The Request** When you click the Spotify icon, that "click" travels through your USB or mouse port. The **chipset** picks up this signal and routes it straight to the **CPU**, telling it: *"Hey, the user wants to open Spotify."*
- **Step 2: Retrieving the Data** The CPU realizes it needs the Spotify program files, which are stored on your Solid State Drive (SSD). The CPU asks for these files, and the **chipset** travels down the data highway to the SSD, grabs the Spotify files,

and helps load them into the system's temporary memory (RAM) so the CPU can read them.

- **Step 3: Processing the App** Now that the files are ready, the **CPU** rapidly processes the code to launch the visual interface of Spotify onto your screen and fetch your personalized playlists.
- **Step 4: Pumping the Sound** When you hit "Play" on a song, the **CPU** decodes the audio file into digital bits. It then hands these bits back to the **chipset**, which routes them precisely to your motherboard's audio chip (or USB headphone jack) so it can be converted into the actual sound you hear in your speakers.

4. **Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.**



diagram

Quick Guide to What You See:

1. **CPU Socket (Large Square):** The central house where the main processor chip gets clamped down to serve as the computer's brain.
2. **RAM Slots (Vertical Lanes):** Long slots placed right next to the CPU for fast-access memory sticks, allowing the system to multi-task smoothly.
3. **PCIe x16 Slot (Long Horizontal Slot):** The heavy-duty expansion slot specifically designed to hold and power your dedicated graphics card (GPU).
4. **M.2 SSD Slot (Small Horizontal Slot):** A compact, high-speed slot tucked directly onto the board surface for lightning-fast solid-state drives.
5. **Rear I/O Ports (Back Edge Panel):** The block of plugs sticking out the back of the computer case where you connect external gear like your mouse, keyboard, monitor, and internet cables.